

**UNIVERSITY OF ILLINOIS-CHICAGO  
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 201 – INTRODUCTION TO THERMODYNAMICS  
Spring 2020**

**Lecture:**

MW: 1:00pm – 2:15pm  
Room: EIB 124

**Office Hours:**

MW 2.15pm-3.15pm (Dr. Bilgin)  
F 10am-12pm (TA)

**Instructor:**

Dr. Betul Bilgin, [bbilgin@uic.edu](mailto:bbilgin@uic.edu)

**Teach Assistants:**

Kun Wang [kwang84@uic.edu](mailto:kwang84@uic.edu)  
Discussion periods: Tuesday/Thursday: 9:30 – 10:20 AM

**Text:**

***Fundamentals of Engineering Thermodynamics*, 9th ed., Moran, Shapiro, Boettner and Bailey (Hardcover, Looseleaf, Ebook can be used)**

**NOTE: MUST USE 9<sup>th</sup> Edition of textbook**

**Course Description:**

1<sup>st</sup> Law of thermodynamics; work, heat and conversion of energy in closed and open systems, including engineering components, power and refrigeration cycles. Engineering problem-solving methodology, pure substance properties, property tables, equations of state, theory of gases and other states of matter. Second Law of thermodynamics; entropy, open and closed system entropy balances and equilibrium, with applications. Fundamental thermodynamic functions, Maxwell relations, Clapeyron equation.

**Course Aims, Scope, Level:**

The engineering science courses that form the basis of chemical engineering can be roughly divided into two sequences: transport processes and thermodynamics. This course is the first in the thermodynamics sequence and is designed to give the student a firm grounding in energy and entropy concepts. Later courses will deal in more detail with phase and reaction equilibrium. Knowledge of thermodynamics is needed in subsequent undergraduate chemical engineering courses. If a student can master the fundamentals early in the course of study, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have the necessary background in elementary mathematics, and this mathematical background will be used. Also, students should understand mechanics which is usually taught in the first college physics course.

By the end of the term, the students will understand energy and entropy concepts for closed systems and open systems (control volumes). Student will be able to apply engineering methodology to a variety of situations in order to solve thermodynamic problems. Energy and entropy are conceptually difficult and require time and effort on the student's part before they are well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as possible, and thoughtful questions should be brought up during lecture and discussion. Good luck!

### **Learning Objectives :**

At the end of the course, students should be able to:

- Apply energy balances for closed systems and control volumes.
- Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
- Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
- Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature-specific volume, temperature-entropy charts.
- Relate thermodynamic properties via partial derivate, Maxwell's relations.
- Interpret phase diagrams of binary systems.
- Apply the entropy balance and identify if the process is reversible or irreversible.

### **Course Outline:**

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 4)
- VI. The Second Law of Thermodynamics (Chapter 5)
- VII. Entropy (Chapter 6)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

### **Exams, Homework and Grading**

There will be 4 to 6 quizzes, two midterm exams, and one final exam. All exams are required. There will be two 1 hour 15 minute-lectures per week (Mon-Wed) and one, TA-led discussion session on Tuesday or Thursday. New material will be covered in lecture. Students should stay current with the material and be prepared to answer questions. During the discussion sessions, homework problems will be worked and occasionally new material covered. The teaching assistant is under no obligation to grade homework that is late, illegible, or sloppy. Scoring fewer than half of the total homework grade is grounds for failing the course. The homework is an important exercise necessary for learning the material and should be worked with care.

## **Project**

There will be one semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a calorimeter. Project includes research phase, modeling phase, prototype phase and testing phase. At the end of the semester, design competition will be held among all the groups. All the groups are required to create AdobeSpark webpage for their design. Students will be graded based on understanding of the concept, design decisions, analysis of testing data and design constraints.

## **Attendance**

Students are expected to attend all classes, and attendance is incorporated as 5% of the final grade. Attendance will be taken at random dates and percentage will be calculated based on the absence of the student.

## **Class Activity**

There will be 4 to 5 class activities through the semester. Students will be working with randomly selected groups on the problems given for that day. If any of the groups answers all the questions right, then everyone attended that day will receive extra credit.

## **SaChE safety modules:**

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for CHE 201:

- The importance of Process Safety
- Hazard Recognition

## **Grading Policy:**

Course grades will be weighted according to the following grading formula.

|                      |     |
|----------------------|-----|
| Two exams (20% each) | 40% |
| Final exam           | 20% |
| Quizzes              | 5%  |
| Homework problems    | 15% |
| Attendance           | 5%  |
| Project              | 15% |

The expectation is that all students will do well in this course if they actively participate in the course, do assignments on time and keep up with course materials.

## **Additional Information and Policies**

1. EXAMS will be held during the class period. Makeup exams will not be provided except for health and weather-related emergencies.
2. Cheating on homework, quizzes, and exams will not be tolerated and appropriate action will be taken including (but not limited to) a failed grade or lowering of the final grade earned in the course.
3. Quizzes will be held periodically during the class period. There will be no makeup quizzes.

4. Late Homework Policy: Homework can be turned in 1 day late without excuse for up to 50% credit.
5. Homework/quiz grading: Lowest homework and quiz grade will be dropped.
6. All the Homework will be submitted via GradeScope. Exams and quizzes will be uploaded to GradeScope by instructor and students will be able to access their graded paper via GradeScope.
7. Lecture Mobile Technology Policy will be given in 1<sup>st</sup> Lecture period.